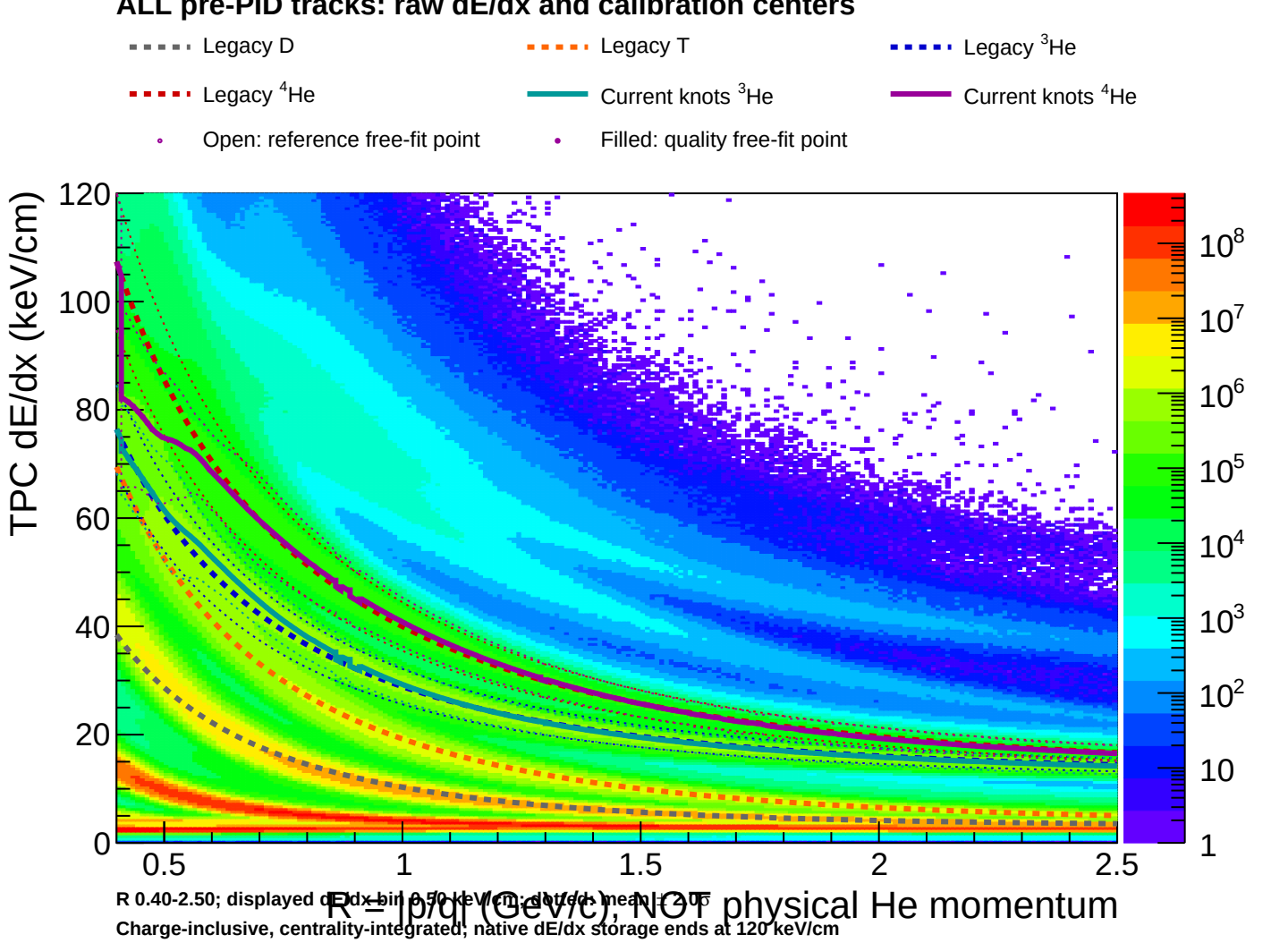
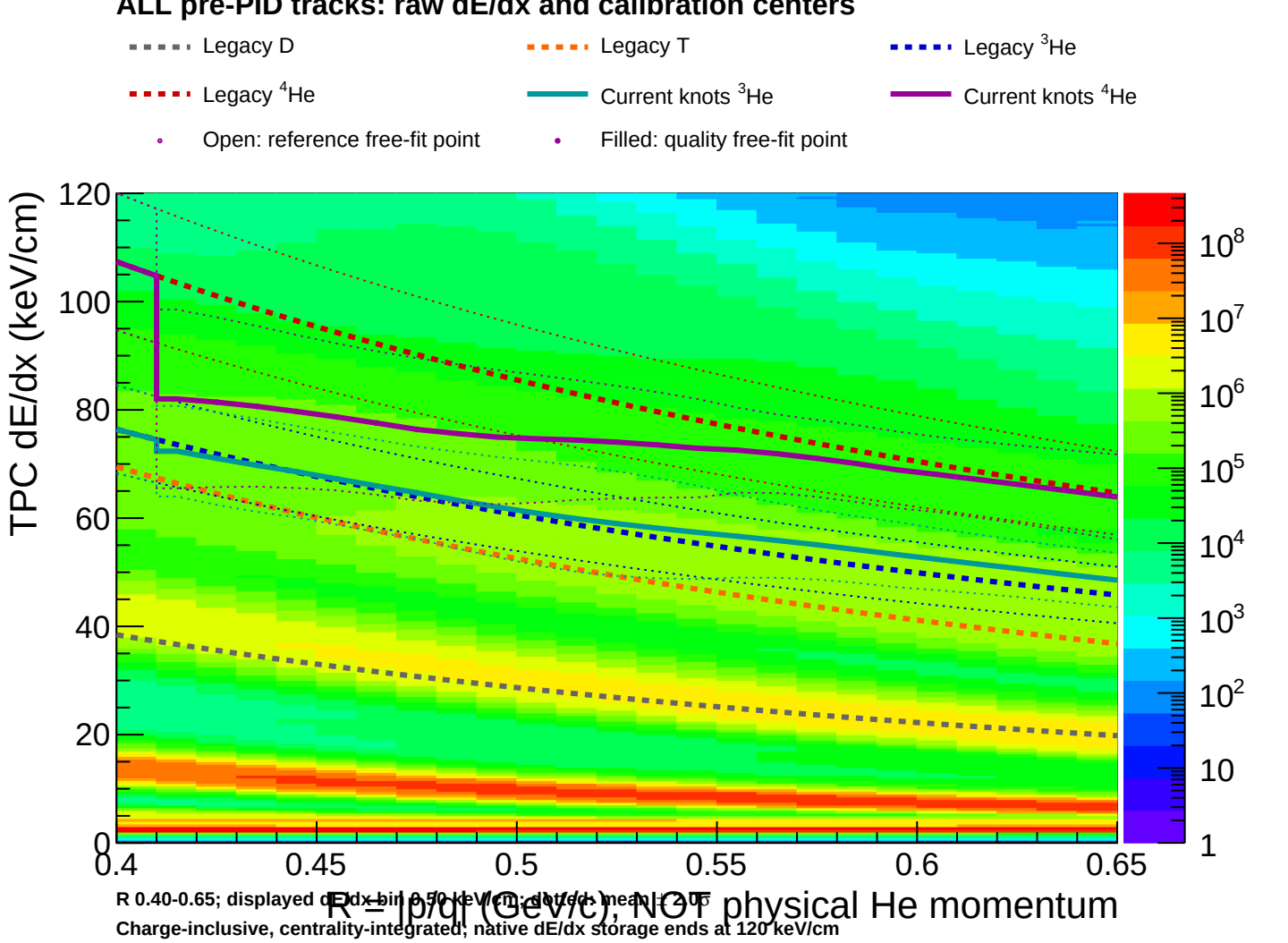
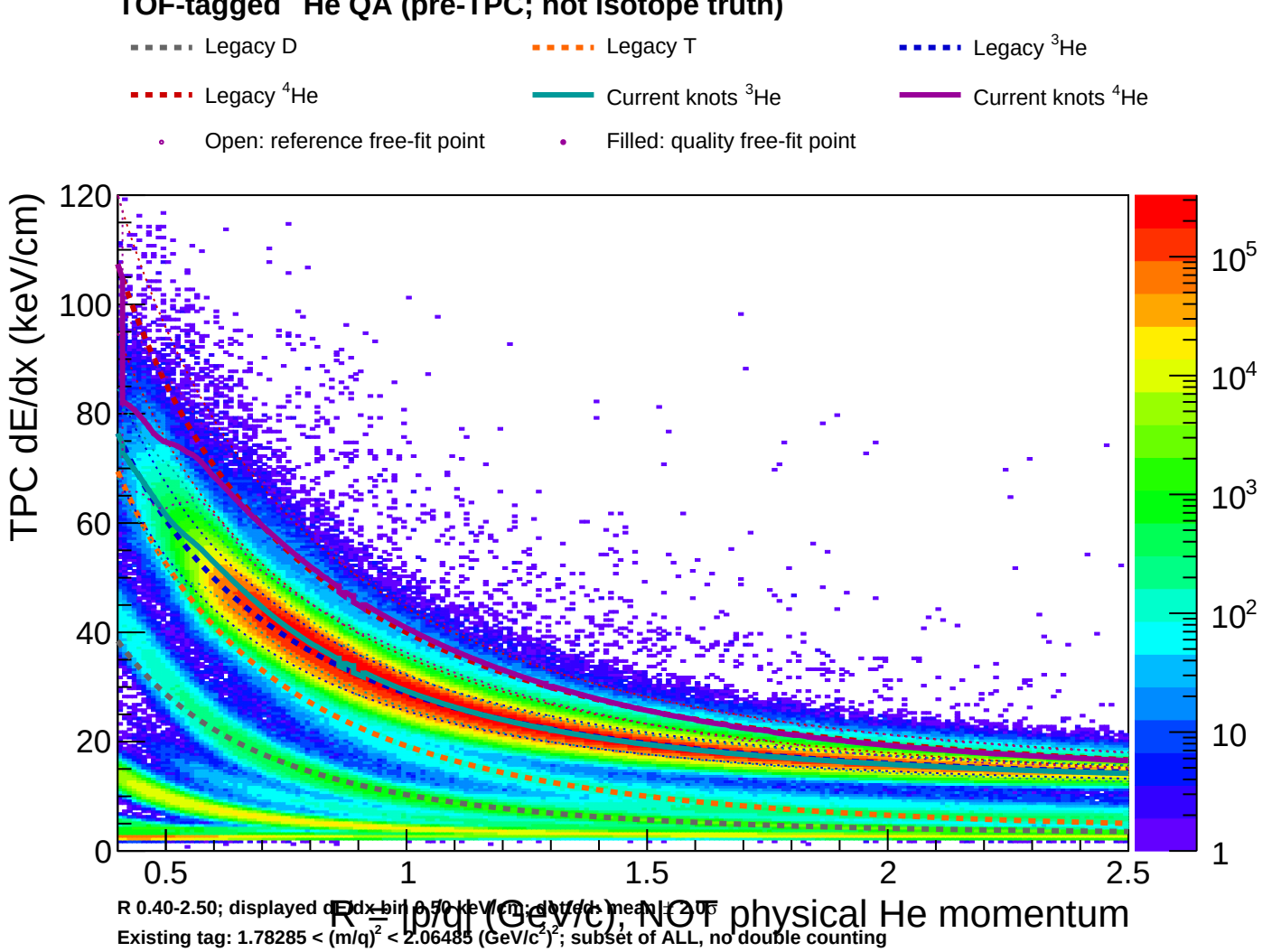
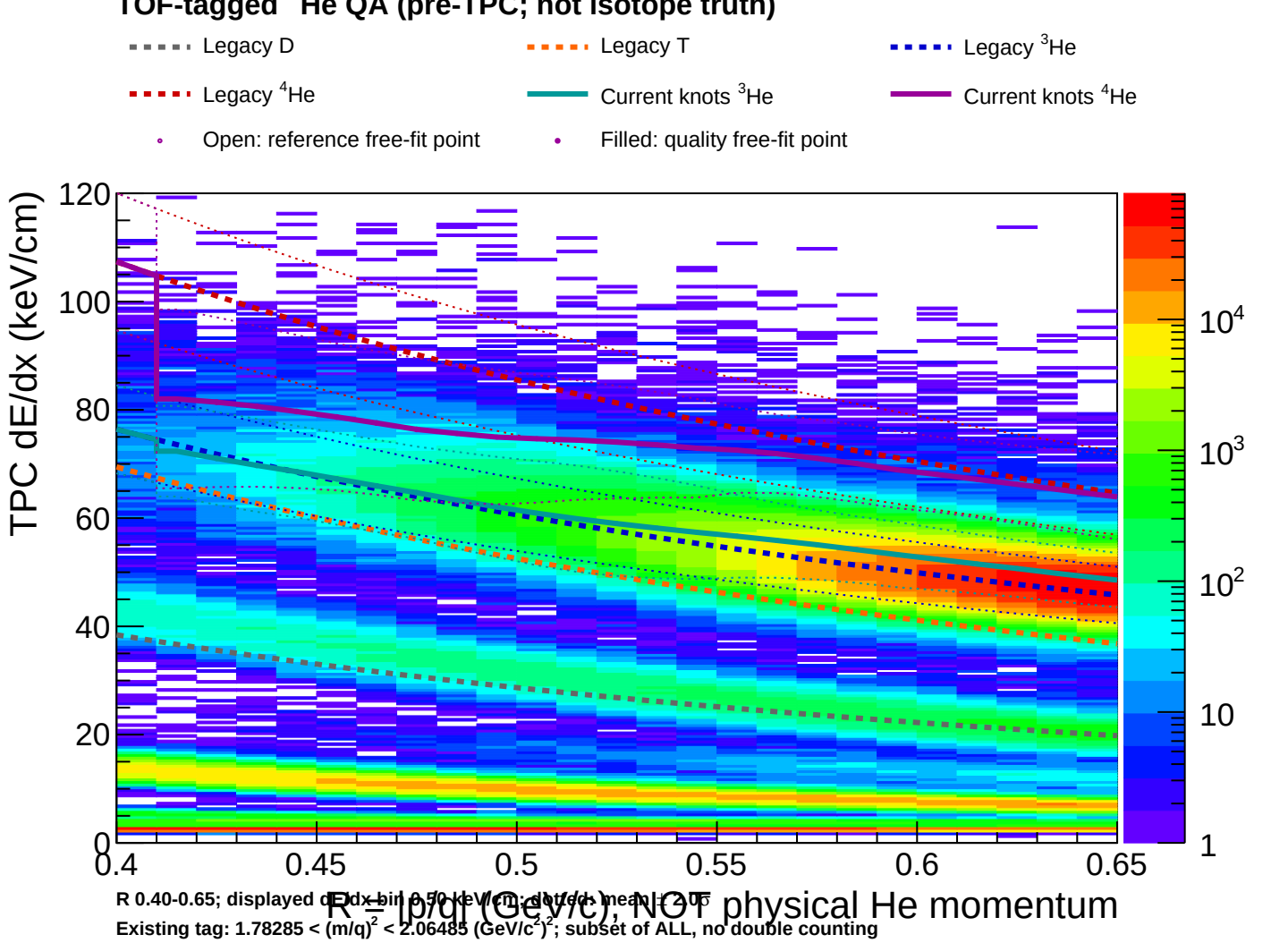

ALL pre-PID tracks: raw dE/dx and calibration centers
Legacy D
Legacy T
Legacy ^{3}He
Legacy ^{4}He
Current knots ^{3}He
Current knots ^{4}He
Open: reference free-fit point
Filled: quality free-fit point
TPC dE/dx (keV/cm)
0
20
40
60
80
100
120
0.5
1
1.5
2
2.5
10^8
10^7
10^6
10^5
10^4
10^3
10^2
10
1
R = |p|q| (GeV/c), NOT physical He momentum
R 0.40-2.50; displayed dE/dx bin 0.50 keV/cm; dotted = mean ± 2 σ
Charge-inclusive, centrality-integrated; native dE/dx storage ends at 120 keV/cm


ALL pre-PID tracks: raw dE/dx and calibration centers
Legacy D
Legacy T
Legacy ^{3}He
Legacy ^{4}He
Current knots ^{3}He
Current knots ^{4}He
Open: reference free-fit point
Filled: quality free-fit point
TPC dE/dx (keV/cm)
120
100
80
60
40
20
0
0.4
0.45
0.5
0.55
0.6
0.65
10^8
10^7
10^6
10^5
10^4
10^3
10^2
10
1
R = |p|/q (GeV/c), NOT physical He momentum
R 0.40-0.65; displayed dE/dx bin 0.50 keV/cm; dotted: mean ± 2 DE
Charge-inclusive, centrality-integrated; native dE/dx storage ends at 120 keV/cm


TOF-tagged ⁴He QA (pre-TPC; not isotope truth)
Legacy D
Legacy T
Legacy ³He
Legacy ⁴He
Current knots ³He
Current knots ⁴He
Open: reference free-fit point
Filled: quality free-fit point
TPC dE/dx (keV/cm)
120
100
80
60
40
20
0
0.5
1
1.5
2
2.5
10⁵
10⁴
10³
10²
10
1
R = |p|/|q| (GeV/c), NOT physical He momentum
R 0.40-2.50; displayed dE/dx bin 0.50 keV/cm; dotted = mean ± 2 RMS
Existing tag: 1.78285 < (m/q)² < 2.06485 (GeV/c²)²; subset of ALL, no double counting


TOF-tagged ³He QA (pre-TPC; not isotope truth)
Legacy D
Legacy T
Legacy ³He
Legacy ⁴He
Current knots ³He
Current knots ⁴He
Open: reference free-fit point
Filled: quality free-fit point
TPC dE/dx (keV/cm)
120
100
80
60
40
20
0
0.4
0.45
0.5
0.55
0.6
0.65
10⁴
10³
10²
10
1
R = |p|/q (GeV/c), NOT physical He momentum
R 0.40-0.65; displayed dE/dx bin 0.50 keV/cm; dotted = mean ± 2 DF
Existing tag: 1.78285 < (m/q)² < 2.06485 (GeV/c²)²; subset of ALL, no double counting